

Try · Compare · Explain

Linear Algebra · Week 1 · 25 September 2026

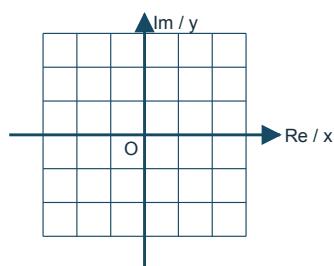
Student worksheet — no solutions | Name (optional): _____

Try each part on your own first, then compare your approach with a partner. Show your working. If stuck, write what you tried and where the difficulty starts. You do not need to finish every activity.

A1 · A complex number as a picture

How many different ways can we represent the complex number $2 - i$?

Start by plotting it and finding its distance from the origin. If you have met polar form, give that too. Label your scale.



Choose and label a suitable scale.

Position, distance, and another representation:

A2 · Vector calculations

Let $\mathbf{u} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. Find $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ where $\mathbf{x} = 3\mathbf{u} - 2(\mathbf{v} + \mathbf{w})$ and $\mathbf{w} - \mathbf{y} = \mathbf{y} - \mathbf{u}$.

(a) Find $\mathbf{x}$.

(b) Find $\mathbf{y}$. Check by substitution.

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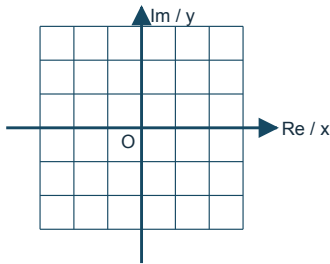
Explain one step to your partner. What did you check?

Directions and coordinates

A3 · Polar to Cartesian

Find $\mathbf{x} \in \mathbb{R}^2$ such that $\mathbf{x} = \lambda \mathbf{u}(\theta)$ for $\lambda = 5$ and $\theta = \pi/4$. Note that $\mathbf{u}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$.

Sketch the direction and length, then calculate both components. Check the length of your answer.

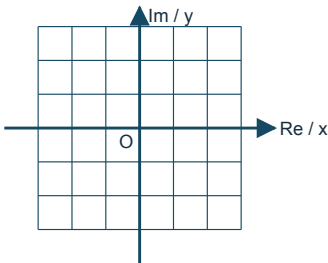


Choose and label a suitable scale.

A4 · Cartesian to polar

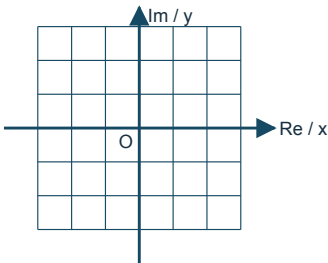
Write $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ in the form $\mathbf{v} = \lambda \mathbf{u}(\theta)$ where $\lambda > 0$, $\theta \in [0, 2\pi)$, and $\mathbf{u}(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$. What about $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$?

Vector (3, 0)



Choose and label a suitable scale.

Vector (0, -3)



Choose and label a suitable scale.

$\lambda =$ _____ $\theta =$ _____

$\lambda =$ _____ $\theta =$ _____

Explain how you chose each angle. Does it satisfy the required range?

Complex arithmetic and a final check

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A5 · Write a complex number in standard form

Write $\frac{(3 + 2i)(3 - 2i)}{i + 1}$ in the form $a + bi$ for $a, b \in \mathbb{R}$.

Simplify the numerator.

Rewrite the fraction in the requested form. Explain why each step preserves its value.

Check your result by multiplying by the original denominator.

A6 · Optional — only if covered in lectures

Compute the dot product $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -8 \end{pmatrix}$ and hence determine the cosine of the angle between the vectors.

Exit check — do this independently.

Write $(0, -2)$ in polar form with a positive radius and an angle in $[0, 2\pi)$. Explain the angle. This is an extra practice check, not a hand-in question.